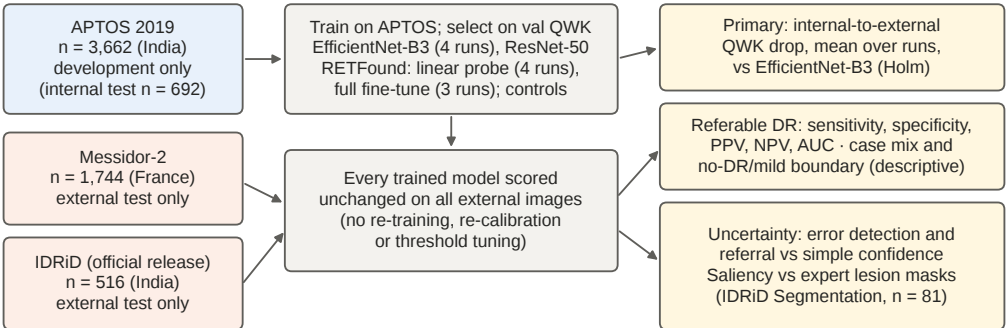
APTOS 2019
n = 3,662 (India)
development only
(internal test n = 692)
Messidor-2
n = 1,744 (France)
external test only
IDRiD (official release)
n = 516 (India)
external test only
Train on APTOS; select on val QWK
EfficientNet-B3 (4 runs), ResNet-50
RETFound: linear probe (4 runs),
full fine-tune (3 runs); controls
Every trained model scored
unchanged on all external images
(no re-training, re-calibration
or threshold tuning)
Primary: internal-to-external
QWK drop, mean over runs,
vs EfficientNet-B3 (Holm)
Referable DR: sensitivity, specificity,
PPV, NPV, AUC · case mix and
no-DR/mild boundary (descriptive)
Uncertainty: error detection and
referral vs simple confidence
Saliency vs expert lesion masks
(IDRiD Segmentation, n = 81)